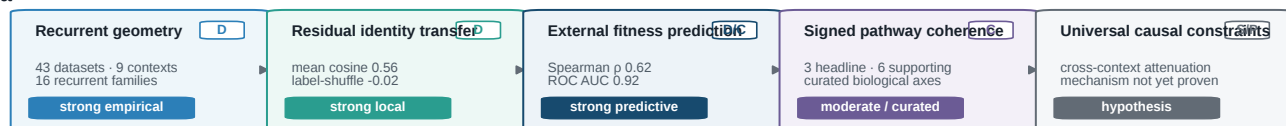


a



b

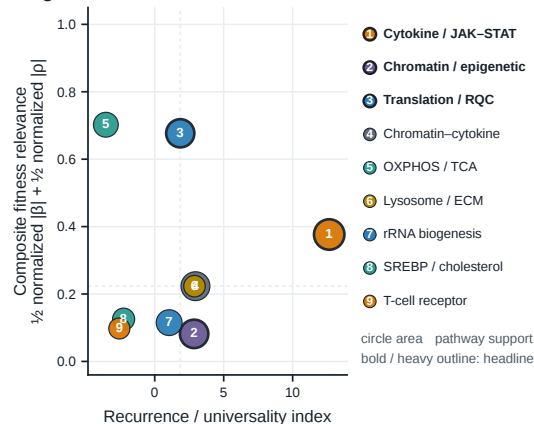
Integrated evidence across curated axes

Cytokine / JAK-STAT	12.65	0.02	0.29	35.3	0.007
Chromatin / epigenetic	2.86	0.03	0.02	22.2	0.026
Translation / RQC	1.85	0.25	0.16	21.4	0.004
Chromatin-cytokine	2.92	0.04	0.13	23.8	0.021
OXPHOS / TCA	-3.54	0.10	0.44	11.3	0.017
Lysosome / ECM	2.92	0.04	0.13	4.4	0.021
rRNA biogenesis	1.05	0.04	0.03	14.0	0.016
SREBP / cholesterol	-2.25	0.04	0.04	5.4	0.013
T-cell receptor	-2.56	0.03	0.04	3.7	0.001
	Recurrence index	Predictive $ \beta $	Essentiality $ \rho $	Pathway $-\log_{10} q$	Coessentiality lift

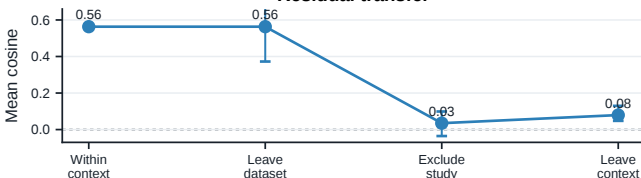
blue stripe: headline teal stripe: supporting

c

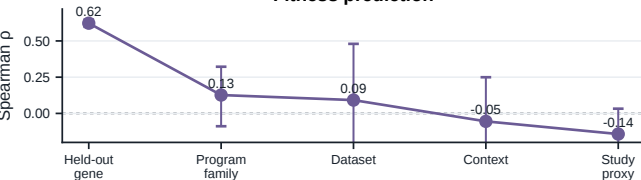
Constraint-axis atlas



Residual transfer

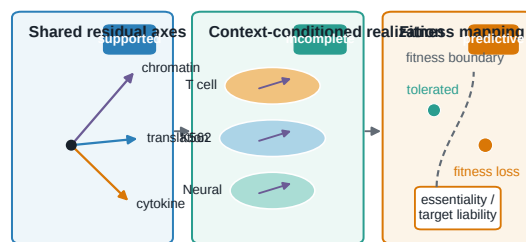


Fitness prediction



e

Hierarchical CGT model



response = shared axes + context modulation + study noise

fitness = $g(\text{axis coordinates, context})$

residual causal law: hypothesis